Certified Robustness for GNN

Nathan Chanak (nchanak@wisc.edu), Diego Navarro (dlnavarro@wisc.edu)

ECE 763 Spring 2026
Group Project

WISCONSIN
UNIVERSITY OF WISCONSIN–MADISON

Motivation

- GNNs are used in sensitive contexts like fraud detection and healthcare analysis
- Small perturbations are enough to change predictions and significantly affect accuracy
- Many empirical defenses can mitigate attacks, but provide no guarantees

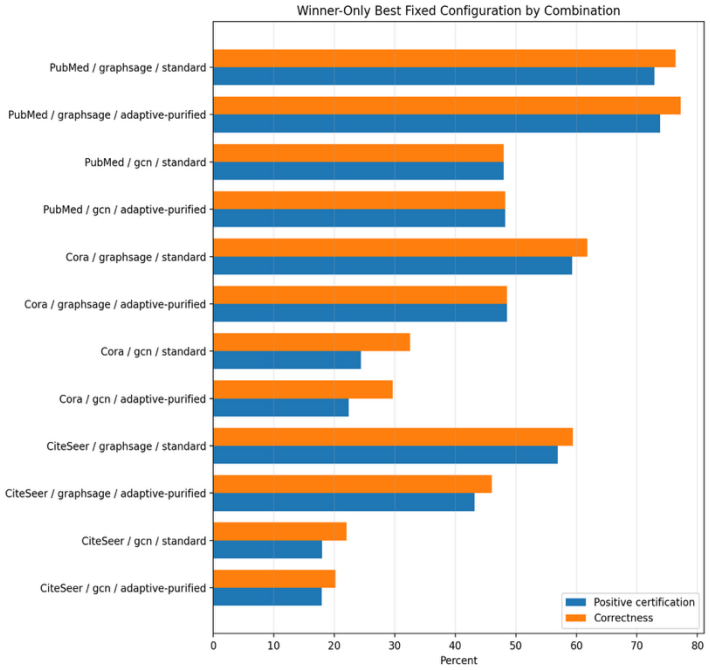
Node	True Label	Clean Prediction	Attacked Prediction	Attack Success
2313	2	2	0	True
1922	0	0	3	True

Techniques

- **Jaccard Purification:** Removes suspicious edges between nodes with dissimilar features
- **Randomized Smoothing:** Samples perturbed versions of the graph and uses majority voting for more certifiable predictions.
- **Sparse Asymmetric Smoothing:** Uses different probabilities for edge deletions and additions, with maximum perturbations, better matching sparse graph attacks where changes are not symmetric.
- **Symmetric Certificates:** Certifies robustness using a single edge flip probability, simpler and often larger radius certificates but with harsher noise.
- **Asymmetric Certificates:** Certifies separate add/delete budgets using likelihood-ratio comparisons
- **Matched Sparse Noisy Training:** Trains with sparse edge flip noise matched to the smoothing setup so the model learns under attack like perturbations.
- **Balanced Sparse Noisy Training:** Uses stronger, more balanced add/delete noise during training to test whether heavier perturbation exposure improves robustness.

Dataset Winners

Dataset	Best Setup	% Nodes Pos. Cert.	% Correct
CiteSeer	GraphSAGE / std.	56.9%	59.4%
Cora	GraphSAGE / std.	59.3%	61.8%
PubMed	GraphSAGE / adaptive	73.8%	77.2%



GraphSAGE is the strongest backbone; PubMed certifies best, while Cora remains the hardest.

Train → Attack → Purify → Smooth → Certify

73.8%
Best positive certification on PubMed / GraphSAGE / adaptive-purified

59.3%
Best positive certification on Cora, achieved by GraphSAGE under standard attack

1.2%
Maximum gap between deployable selector and oracle over all 12 combinations

Challenges

- Graph perturbations are discrete and combinatorial, so image style certificates can become overly conservative on graphs.
- Our local symmetric certificate often collapses to radius 0 even when the smoothed prediction stays correct.

Main bottleneck: The pipeline works end-to-end, but we still need sparse, graph-aware certificates that do not collapse under realistic low-noise edge perturbations.

Next Steps

- Extend the current symmetric certificate to a sparsity aware asymmetric certificate.
- Run larger multi seed studies with wider target pools and stronger uncertainty estimates.
- Learn dataset specific purification and smoothing settings, then test beyond citation graphs.

Immediate follow-up: prioritize the asymmetric-certificate isolation experiments, because that is the clearest path to stronger certified radii without losing current accuracy gains.